CONTROLLED BIAS IN RECOMMENDER SYSTEMS

Fake ratings change real recommendations

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Same synthetic profiles.
Different SVD and NMF responses.

THE BOOK RESULT

At 500 fake profiles, NMF recommends far more Thriller.

Mean target books in each real user's Top-15 list

SVD 2.28 / 15
15.2% of recommendation slots

NMF 10.84 / 15
72.3% of recommendation slots

BOOKS — 13 assigned genres
53,424 real users
9,415 items | 5,815,569 ratings

MOVIES — Release-decade categories
943 real users
1,682 items | 100,000 ratings

MOBILE APPS — 48 original categories
700,111 real users
10,173 items | 19,297,019 ratings

HOW THE EXPERIMENT WORKS

01 Create fake profiles
Target items = 5
All other items = 1

→

02 Append their ratings
Keep real users, ratings
and the catalog unchanged

→

03 Train both models
Biased Surprise SVD / NMF
8 factors; 180 epochs; seed 42

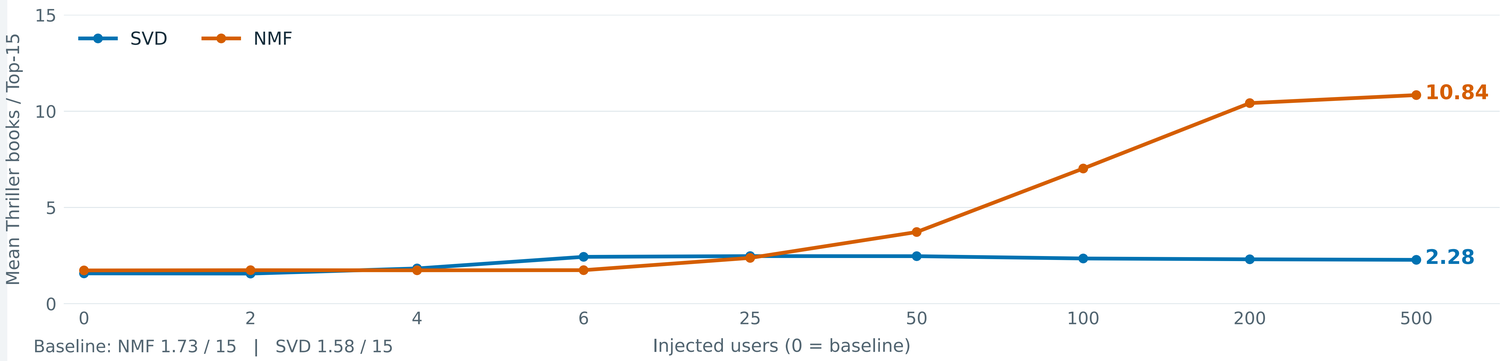
→

04 Measure real-user output
Rank unseen items
Count target items in Top-15

Dense profiles: 500 fake book users add 4,707,500 ratings - 80.9% of the original book rating count.

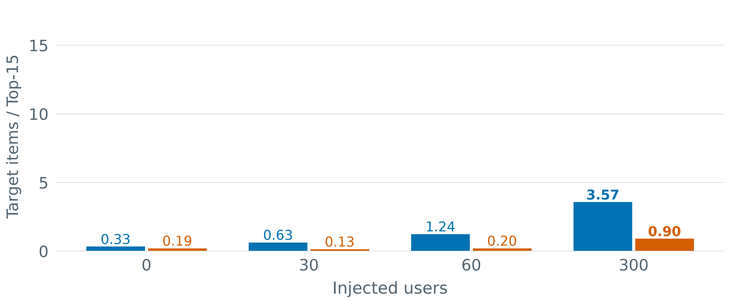
01 BOOKS / THRILLER

The response diverges as more identical profiles are injected.

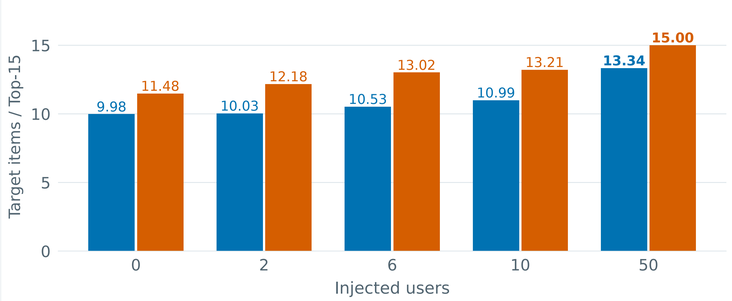


Baseline: NMF 1.73 / 15 | SVD 1.58 / 15

02 MOBILE APPS / SHOPPING



03 MOBILE APPS / 1990S



SAME PREDICTION FORMULA

$$\hat{r}_{ui} = \mu + b_u + b_i + p_u^\top q_i$$

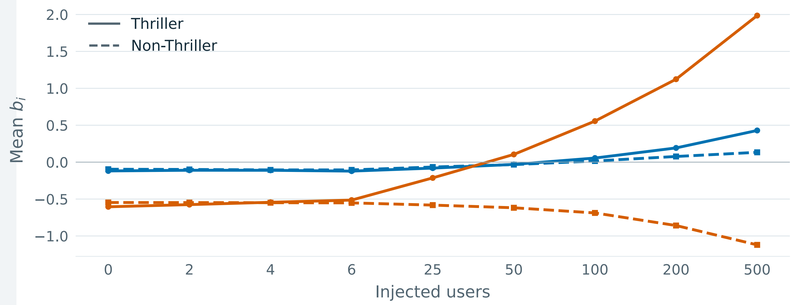
Global mean + user bias + book bias + user-item dot product.

SVD Signed latent factors; additive gradient updates.

NMF Non-negative factors; multiplicative factor updates. Biases can be negative.

WHAT CHANGES IN BOOK BIAS?

Thriller injection; equal weight per book in this mean



NMF develops a larger book-bias separation between target and non-target books, alongside stronger Thriller exposure.

TAKEAWAY

Identical synthetic ratings can produce very different exposure shifts.
The response depends on the model, target category and injection budget.

SCOPE Exposure and fitted parameters, not held-out accuracy. Movie target has high baseline exposure; user counts and rating budgets differ across domains.

DATA: Goodbooks-10K | MovieLens 100K | MobileRec. Book genre labels assigned with GPT-3.5.

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